

Result Reproduction

1 Goncalves et al., “Numerical solution of the time dependent Ginzburg-Landau equations for mixed (d + s)- wave superconductors”

We try to reproduce section VIII, the paper’s computational results. The parameters are $\kappa_D = 2$, $\tau_1 = 8/3$, $\tau_3 = 16/3$, $\tau_4 = 2$, $\nu = -2$, $\eta_s = 2$, $\eta_v = 1$. Note that these correspond to microscopically derived coefficients from other papers. The square film has dimensions $8\xi_D \times 8\xi_D$. The grid starts off with (everywhere) $\psi_D = 1$, $\psi_S = 0$, $\mathbf{B} = 0$. In the paper, they gradually sweep the field up to $2.5H_{c2}$ and back down to 0. My computer only had the capability to step in intervals of $0.5H_{c2}$, which probably accounts for the differences in behavior below. Another possibility is that solving on a mesh rather than a grid causes differences, but I think that’s less likely.

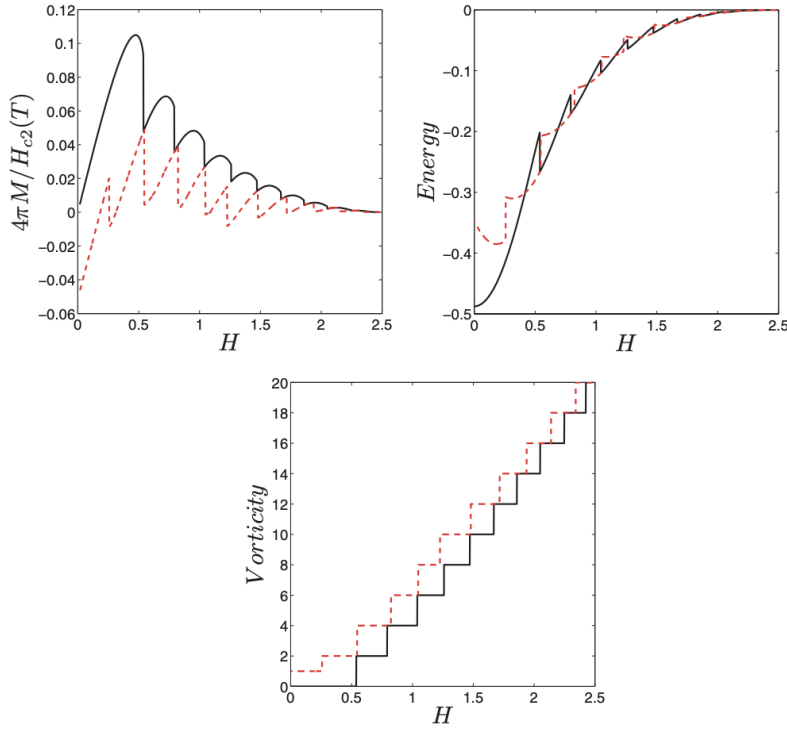


FIG. 2. The magnetization (upper-left) and the Gibbs free-energy (upper-right) as a function of the external applied field; for magnetic field sweeping up (black curve) and then down (red dashed curve). Different states are labeled by their corresponding vorticity L (lower panel).

Figure 1: Field dependence reported by Goncalves et al.

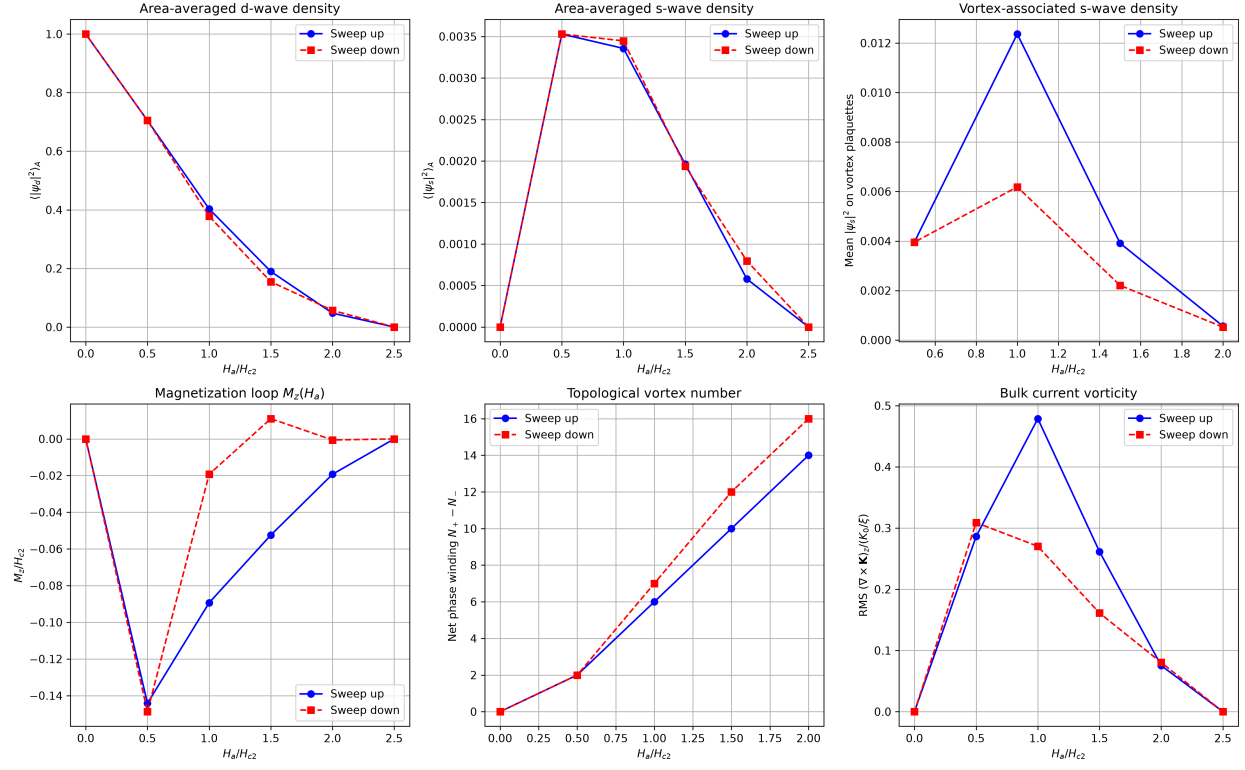


Figure 2: Reproduced field dependence using a sweep interval of $0.5H_{c2}$.

One issue, among many other differences (due to my intervals), is that there is no vortex remaining when the system gets back to $H = 0$. However, the vortex count seems to roughly track otherwise.

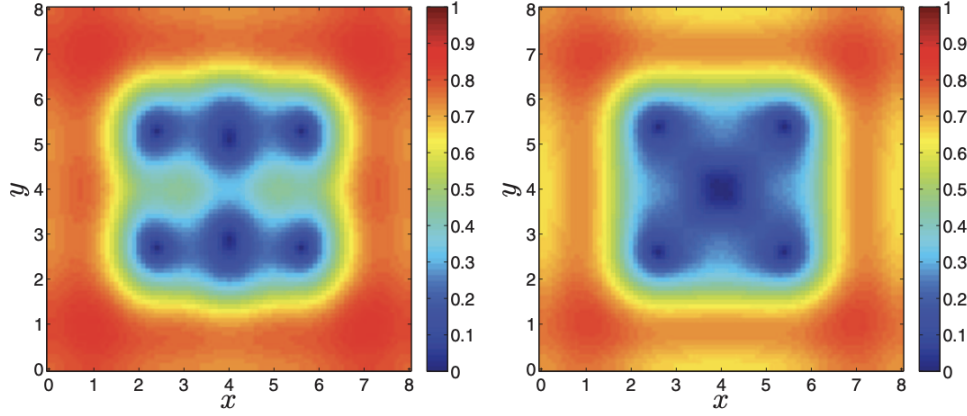


FIG. 6. Density plot of $|\psi_d|$ for six vortices at $H = 1.040$ (left) and $H = 1.125$ (right).

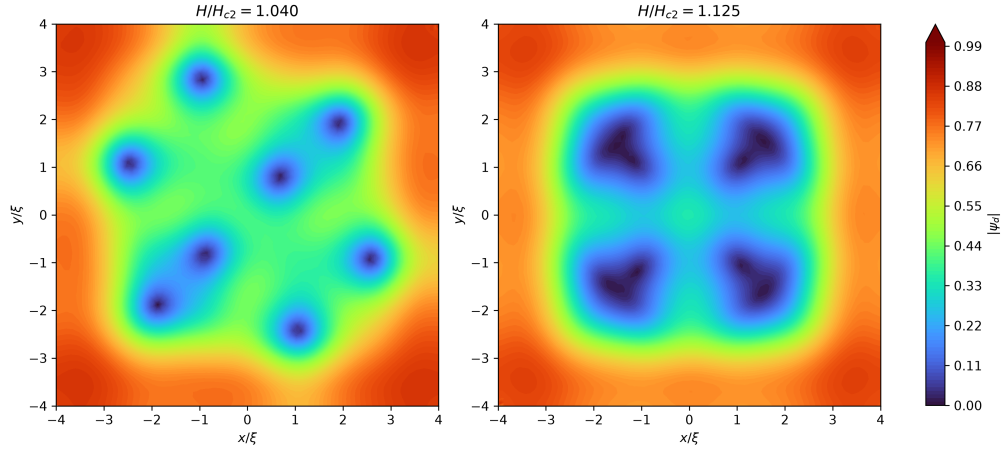


Figure 3: Reference (top) and reproduced (bottom) d -wave vortex configurations near $H/H_{c2} = 1.1$.

I tried to reproduce the plots in Goncalves's figure 6 by going straight from $H = 0$ to the desired values, but again hysteresis seems to cause behavioral differences (in mine there are 8 vortices rather than 6).

2 Xu, Ren, Ting, “Structures of single vortex and vortex lattice in a d-wave superconductor”

The single-vortex magnitude and magnetic-field profiles reported by Xu, Ren, and Ting are shown below, followed by their phase textures for the dominant d -wave and induced s -wave components.

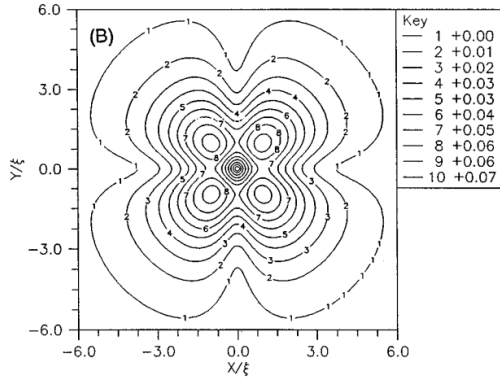


FIG. 1. Distribution of $|\psi_s|$ around a single vortex for $T/T_c=0.5$, $\alpha_s=1$, and $\kappa=2$: (A) surface plot and (B) contour plot.

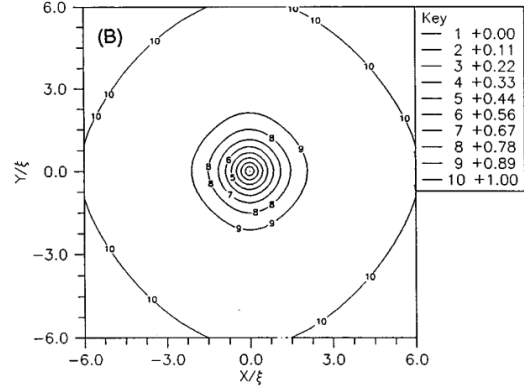


FIG. 2. Same as Fig. 1 but for $|\psi_d|$.

Figure 4: Reference single-vortex magnitude profiles for $|\psi_s|$ (left) and $|\psi_d|$ (right).

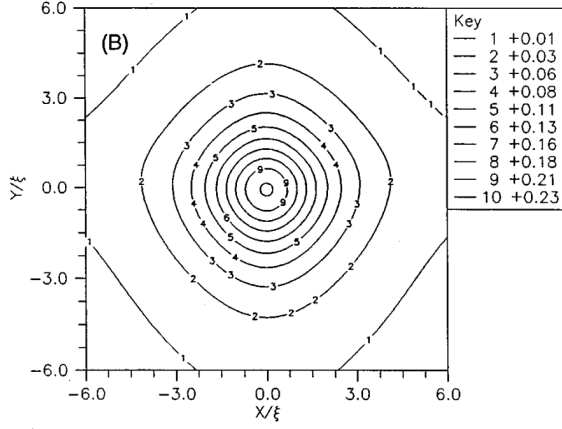


FIG. 3. Same as Fig. 1 but for h .

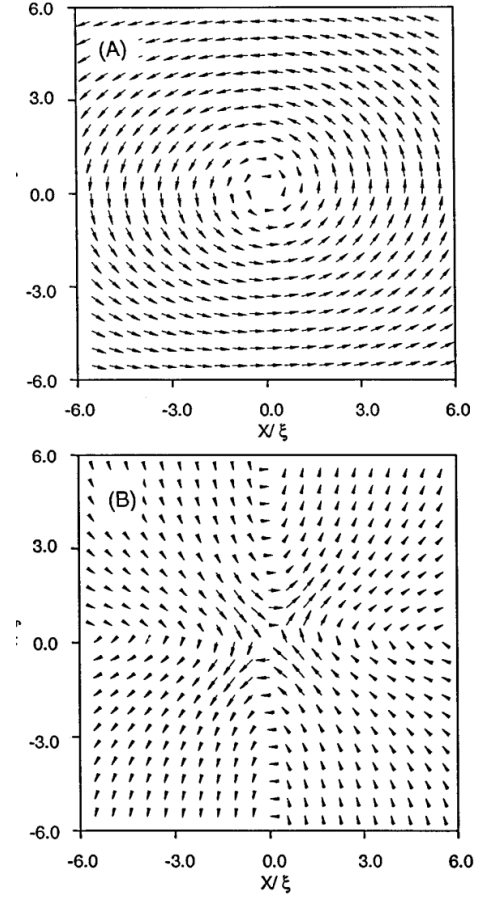


FIG. 4. Phases of ψ_d (A) and ψ_s (B).

Figure 5: Reference single-vortex magnetic-field profile (left) and phase textures of ψ_d and ψ_s (right).

The corresponding simulation using the Goncalves parameters, a centered circular defect of radius 0.5ξ , and $H/H_{c2} = 0.5$ is shown below. The level curves show the order-parameter magnitudes and magnetic fields, while the arrows show the phases of ψ_d and ψ_s .

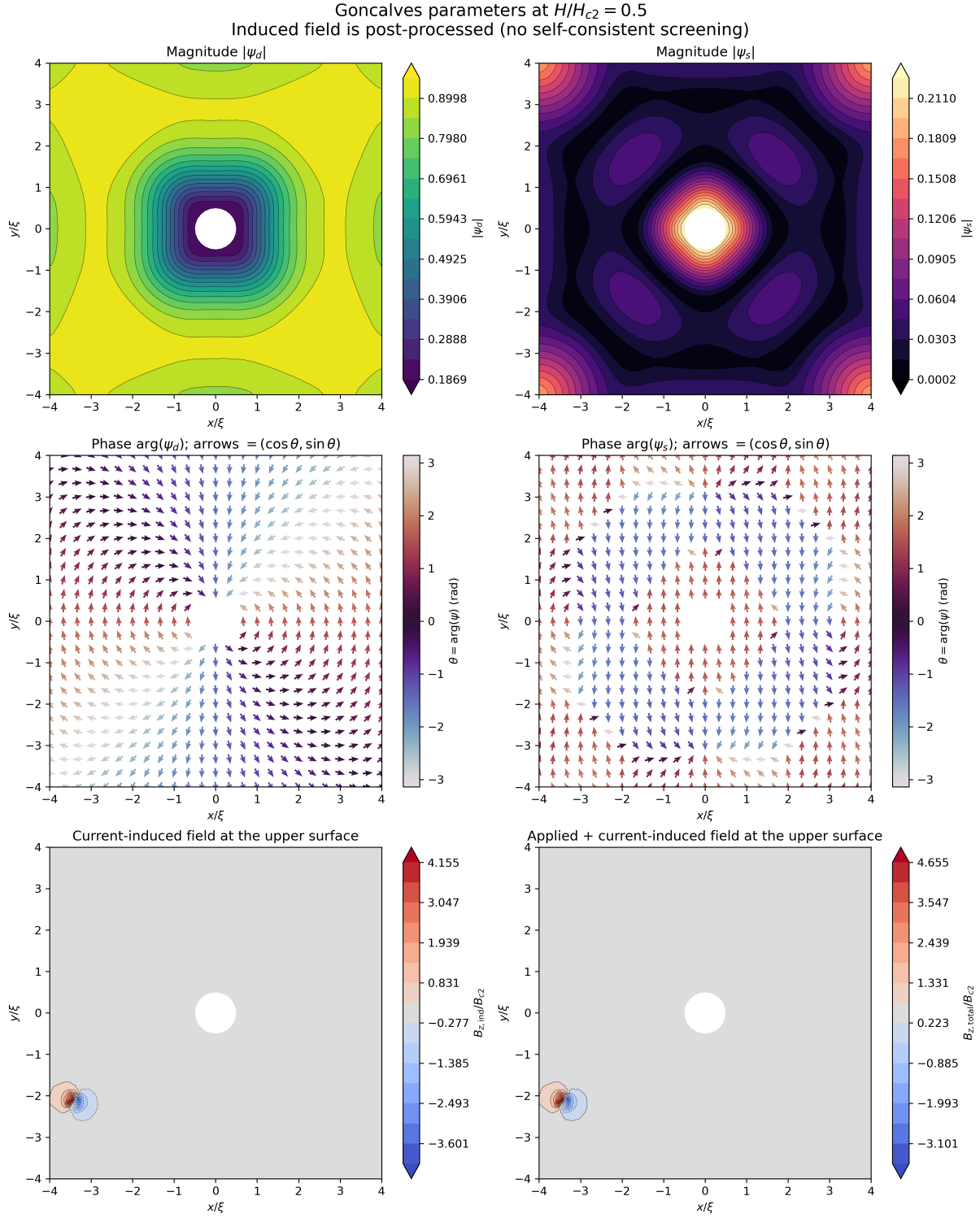


Figure 6: Xu–Ren–Ting-inspired defect simulation at $H/H_{c2} = 0.5$.

- 3 Lin et al., “Distinguishing between s+id and s+is pairing symmetries in multiband superconductors through spontaneous magnetization pattern induced by a defect”
- 4 Lei et al., “Field Driven Pairing State Phase Transition in $d_{x^2-y^2} + id_{xy}$ -Wave Superconductors”